



PRODUCT NAME

Premium Organic Shower Gel Formula

Description

A high-quality, cost-effective shower gel formulation with cleansing, pH-balanced, and biodegradable properties. Designed to provide a refreshing and eco-friendly bathing experience.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	18 months
Batch Size:	100 - 200 kg
Processing Time:	3 hours
Temperature:	Mix at 40°C and cool to below 30°C before adding sensitive components.
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Main mixing vessel, auxiliary mixing container, heating element, calibrated pH meter
Certification:	Suitable for natural product certification

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	15.0%	Cleansing agent
3	Decyl Glucoside	Decyl Glucoside	10.0%	Mild surfactant
4	Glycerin	Glycerin	5.0%	Humectant
5	Xanthan Gum	Xanthan Gum	1.0%	Viscosity modifier
6	Citric Acid	Citric Acid	1.0%	pH adjuster
7	Preservative Eco	Benzyl Alcohol, Salicylic Acid, Glycerin, Sorbic Acid	1.0%	Preservative
8	Essential Oil Blend	Fragrance	1.0%	Fragrance
9	Aloe Vera Extract	Aloe Barbadensis Leaf Juice	5.0%	Moisturizing agent
10	Sodium Chloride	Sodium Chloride	1.0%	Viscosity adjuster
Total			100%	

Manufacturing Process

Phase 1: Aqueous Phase

1. In a main mixing vessel, combine water, glycerin, and xanthan gum.
2. Heat to 40°C while stirring until the mixture is homogeneous.

Phase 2: Surfactant Phase

1. In a separate container, mix cocamidopropyl betaine and decyl glucoside.
2. Add to the aqueous phase slowly while mixing.

Phase 3: Adjustment Phase

1. Slowly add citric acid to adjust the pH to 7, monitoring with a calibrated pH meter.
2. Add sodium chloride to reach desired viscosity and continue to mix.

Phase 4: Cooling and Preservation Phase

1. Cool the mixture to below 30°C.
2. Incorporate preservative and essential oil blend, mixing gently.
3. Lastly, add aloe vera extract and mix until uniform.

Required Equipment

Main mixing vessel, auxiliary mixing container, heating element, calibrated pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Suitable for natural product certification

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100 - 200 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount onto a wet bath sponge or washcloth, lather, and rinse off thoroughly.
Suitable for daily use.